

La base des bascules

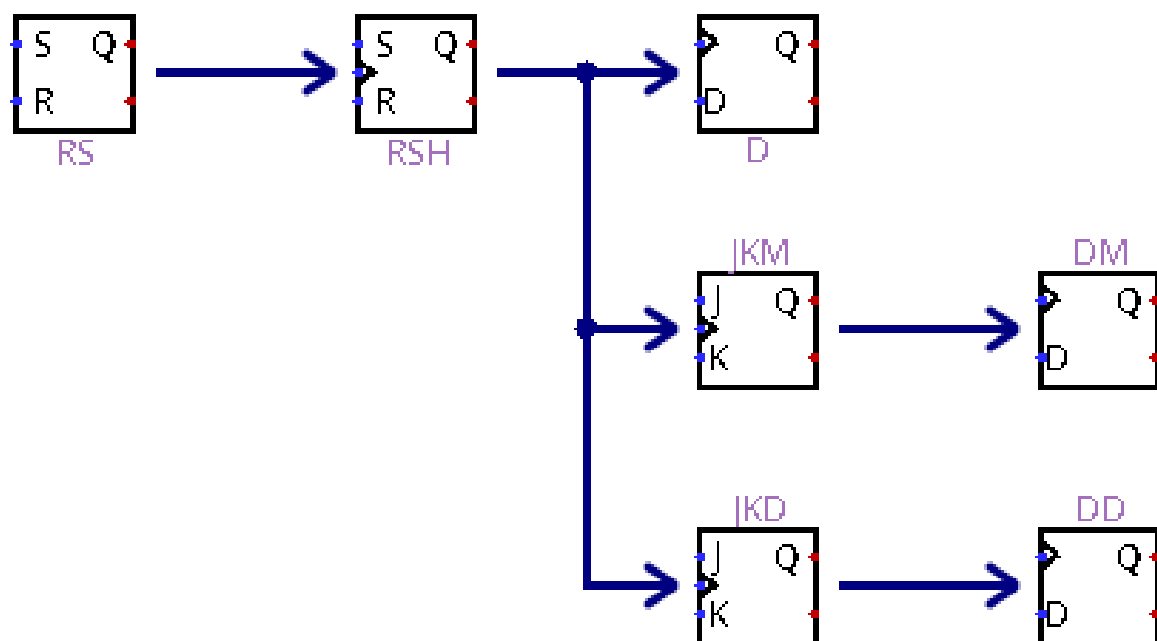
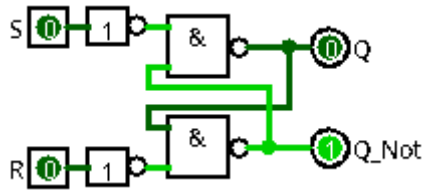


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1) Bascule RS :

1-1) NON ET : (RS1)



$$Q = Q_{n-1} \cdot \bar{R} + S$$

$$\bar{Q} = \bar{Q}_{n-1} \cdot \bar{S} + R$$

S	R	Q	$\bar{Q}$	remarque
0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	1	mise à 0
1	0	1	0	mise à 1
1	1	X	X	état interdit

Algèbre de boole

$$Q = \overline{\bar{Q} \cdot \bar{S}}$$

$$\bar{Q} = \overline{Q \cdot \bar{R}}$$

$$Q = \overline{(\bar{Q} \cdot \bar{R}) \cdot \bar{S}}$$

$$Q = \overline{(\bar{Q} + \bar{R}) \cdot \bar{S}}$$

$$Q = \overline{(\bar{Q} + R) \cdot \bar{S}}$$

$$Q = \overline{\bar{Q} \cdot \bar{S} + R \cdot \bar{S}}$$

$$Q = \overline{(\bar{Q} \cdot \bar{S}) + (R \cdot \bar{S})}$$

$$Q = \overline{(\bar{Q} \cdot \bar{S}) \cdot (R \cdot \bar{S})}$$

$$Q = \overline{(\bar{Q} + \bar{S}) \cdot (\bar{R} + \bar{S})}$$

$$Q = \overline{(Q + S) \cdot (\bar{R} + S)}$$

$$Q = \overline{(Q + S) \cdot (\bar{R} + S)}$$

$$Q = Q \cdot \bar{R} + Q \cdot S + S \cdot \bar{R} + S \cdot S$$

$$Q = Q \cdot \bar{R} + S \cdot Q + S \cdot \bar{R} + S \cdot 1$$

$$Q = Q \cdot \bar{R} + S \cdot (Q + \bar{R} + 1)$$

$$Q = Q \cdot \bar{R} + S$$

$$Q = Q_{n-1} \cdot \bar{R} + S$$

$$\bar{Q} = \overline{(\bar{Q} \cdot \bar{S}) \cdot \bar{R}}$$

$$\bar{Q} = \overline{(\bar{Q} + \bar{S}) \cdot \bar{R}}$$

$$\bar{Q} = \overline{(Q + S) \cdot \bar{R}}$$

$$\bar{Q} = \overline{Q \cdot \bar{R} + S \cdot \bar{R}}$$

$$\bar{Q} = \overline{(Q \cdot \bar{R}) + (S \cdot \bar{R})}$$

$$\bar{Q} = \overline{(\bar{Q} + \bar{R}) \cdot (\bar{S} + \bar{R})}$$

$$\bar{Q} = \overline{(\bar{Q} + \bar{R}) \cdot (\bar{S} + R)}$$

$$\bar{Q} = \overline{(\bar{Q} + R) \cdot (\bar{S} + R)}$$

$$\bar{Q} = \overline{(\bar{Q} + R) \cdot (\bar{S} + R)}$$

$$\bar{Q} = \bar{Q} \cdot \bar{S} + \bar{Q} \cdot R + R \cdot \bar{S} + R \cdot R$$

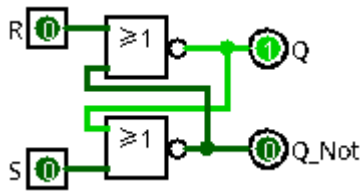
$$\bar{Q} = \bar{Q} \cdot \bar{S} + \bar{Q} \cdot R + R \cdot \bar{S} + 1 \cdot R$$

$$\bar{Q} = \bar{Q} \cdot \bar{S} + R \cdot (\bar{Q} + \bar{S} + 1)$$

$$\bar{Q} = \bar{Q} \cdot \bar{S} + R$$

$$\bar{Q} = \bar{Q}_{n-1} \cdot \bar{S} + R$$

1-2) NON OU : (RS)



$$Q = Q_{(n-1)} \cdot \bar{R} + S$$

$$\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{S} + R$$

S	R	Q	$\bar{Q}$	remarque
0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	1	mise à 0
1	0	1	0	mise à 1
1	1	X	X	état interdit

Algèbre de boole

$$Q = \overline{\bar{Q} + R}$$

$$\bar{Q} = \overline{Q + S}$$

$$Q = \overline{\bar{Q} + S + R}$$

$$Q = (\bar{Q} \cdot \bar{S}) + R$$

$$Q = (\bar{Q} + R) \cdot (\bar{S} + R)$$

$$Q = \bar{Q} \cdot \bar{S} + \bar{Q} \cdot R + R \cdot \bar{S} + R \cdot R$$

$$Q = \bar{Q} \cdot \bar{S} + \bar{Q} \cdot R + R \cdot \bar{S} + R \cdot 1$$

$$Q = R \cdot (\bar{Q} + \bar{S} + 1) + \bar{Q} \cdot \bar{S}$$

$$Q = R + \bar{Q} \cdot \bar{S}$$

$$Q = \bar{R} \cdot Q + S$$

$$Q = Q \cdot \bar{R} + S$$

$$Q = Q_{(n-1)} \cdot \bar{R} + S$$

$$\bar{Q} = \overline{\bar{Q} + R + S}$$

$$\bar{Q} = (\bar{Q} \cdot \bar{R}) + S$$

$$\bar{Q} = (\bar{Q} + S) \cdot (\bar{R} + S)$$

$$\bar{Q} = \bar{Q} \cdot \bar{R} + \bar{Q} \cdot S + S \cdot \bar{R} + S \cdot S$$

$$\bar{Q} = \bar{Q} \cdot \bar{R} + \bar{Q} \cdot S + S \cdot \bar{R} + S \cdot 1$$

$$\bar{Q} = S \cdot (\bar{Q} + \bar{R} + 1) + \bar{Q} \cdot \bar{R}$$

$$\bar{Q} = S + \bar{Q} \cdot \bar{R}$$

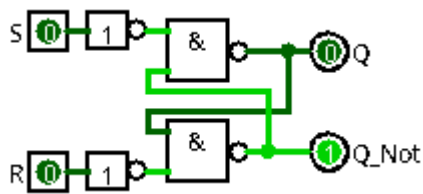
$$\bar{Q} = \bar{S} \cdot \bar{Q} + R$$

$$\bar{Q} = \bar{Q} \cdot \bar{S} + R$$

$$\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{S} + R$$

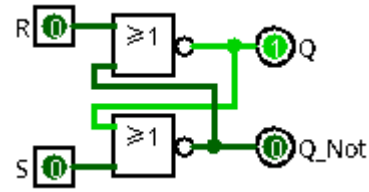
1-3) Résultat :

R=0 S=0	R=1 S=0	R=0 S=1	R=1 S=1
$R=0$ $S=0$ $Q = Q_{(n-1)} \cdot \bar{0} + 0$ $Q = Q_{(n-1)} \cdot 1$ $Q = Q_{(n-1)}$ $\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{0} + 0$ $\bar{Q} = \bar{Q}_{(n-1)} \cdot 1$ $\bar{Q} = \bar{Q}_{(n-1)}$	$R=1$ $S=0$ $Q = Q_{(n-1)} \cdot \bar{1} + 0$ $Q = Q_{(n-1)} \cdot 0$ $Q = 0$ $\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{0} + 1$ $\bar{Q} = \bar{Q}_{(n-1)} \cdot 1 + 1$ $\bar{Q} = \bar{Q}_{(n-1)} + 1$ $\bar{Q} = 1$	$R=0$ $S=1$ $Q = Q_{(n-1)} \cdot \bar{0} + 1$ $Q = Q_{(n-1)} \cdot 1 + 1$ $Q = Q_{(n-1)} + 1$ $Q = 1$ $\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{1} + 0$ $\bar{Q} = \bar{Q}_{(n-1)} \cdot 0$ $\bar{Q} = 0$	$R=1$ $S=1$ $Q = Q_{(n-1)} \cdot \bar{1} + 1$ $Q = Q_{(n-1)} \cdot 0 + 1$ $Q = 1$ $\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{1} + 1$ $\bar{Q} = \bar{Q}_{(n-1)} \cdot 0 + 1$ $\bar{Q} = 1$
Mémoire	Mise à 0	Mise à 1	Interdit



$$\begin{aligned} Q &= Q_{n-1} \cdot \bar{R} + S \\ \bar{Q} &= \overline{Q_{n-1}} \cdot \bar{S} + R \end{aligned}$$

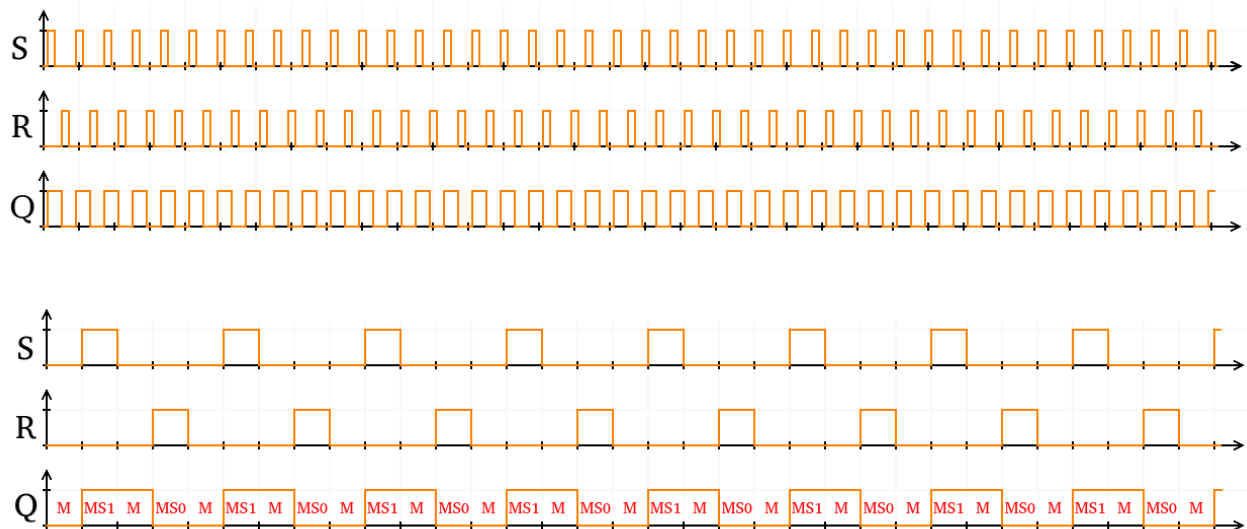
S	R	Q	$\overline{Q}$	remarque
0	0	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	1	0	1	mise à 0
1	0	1	0	mise à 1
1	1	X	X	état interdit



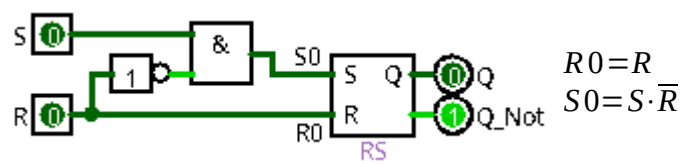
$$\begin{aligned} Q &= Q_{(n-1)} \cdot \bar{R} + S \\ \bar{Q} &= \overline{Q_{(n-1)}} \cdot \bar{S} + R \end{aligned}$$

S	R	Q	$\bar{Q}$	remarque
0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	1	mise à 0
1	0	1	0	mise à 1
1	1	X	X	état interdit

1-4) diagramme de temps :



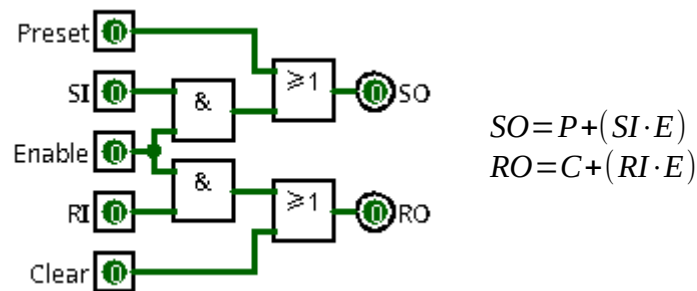
1-5) Bascule Logisim : (RS => RSL)



R	S	R0	S0
0	0	0	0
0	1	0	1
1	0	1	0
1	1	1	0

2) Preset Enable Clear :

2-1) PECM : (Preset Enable Clear Main)



E=0 P=0 C=0

$SO=0$
 $RO=0$

$SO=0+(SI \cdot 0)$
 $SO=0+0$
 $SO=0$

$RO=0+(RI \cdot 0)$
 $RO=0+0$
 $RO=0$

Mémoire

E=1 P=0 C=0

$SO=SI$
 $RO=RI$

$SO=0+(SI \cdot 1)$
 $SO=0+SI$
 $SO=SI$

$RO=0+(RI \cdot 1)$
 $RO=0+RI$
 $RO=RI$

Bascule RS

E=0 P=1 C=0

$SO=1$
 $RO=0$

$SO=1+(SI \cdot 0)$
 $SO=1+0$
 $SO=1$

$RO=0+(RI \cdot 0)$
 $RO=0+0$
 $RO=0$

Mise à 1

E=0 P=0 C=1

$SO=0$
 $RO=1$

$SO=0+(SI \cdot 0)$
 $SO=0+0$
 $SO=0$

$RO=1+(RI \cdot 0)$
 $RO=1+0$
 $RO=1$

Mise à 0

E=0 P=1 C=1

$SO=1$
 $RO=1$

$SO=1+(SI \cdot 0)$
 $SO=1+0$
 $SO=1$

$RO=1+(RI \cdot 0)$
 $RO=1+0$
 $RO=1$

Interdit

E=1 P=1 C=1

$SO=1$
 $RO=1$

$SO=1+(SI \cdot 1)$
 $SO=1+SI$
 $SO=1$

$RO=1+(RI \cdot 1)$
 $RO=1+RI$
 $RO=1$

Interdit

E=1 P=0 C=1

$SO=SI$
 $RO=1$

$SO=0+(SI \cdot 1)$
 $SO=0+SI$
 $SO=SI$

$RO=1+(RI \cdot 1)$
 $RO=1+RI$
 $RO=1$

SI=0 : Mise à 0
SI=1 : Interdit

E=1 P=1 C=0

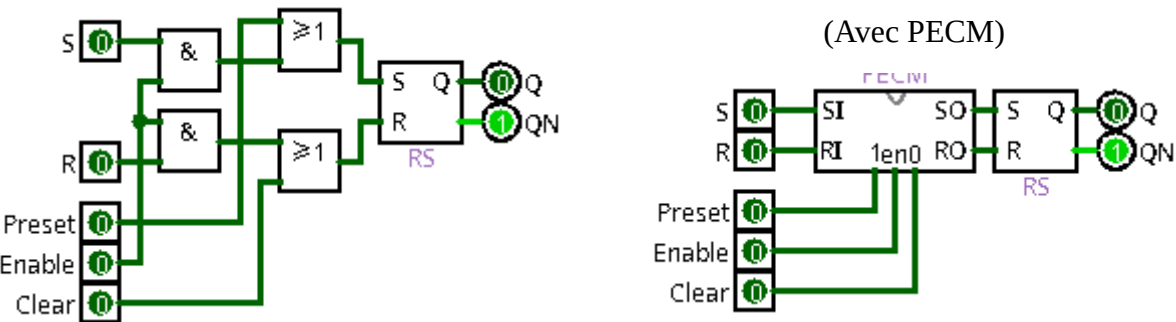
$SO=1$
 $RO=RI$

$SO=1+(SI \cdot 1)$
 $SO=1+SI$
 $SO=1$

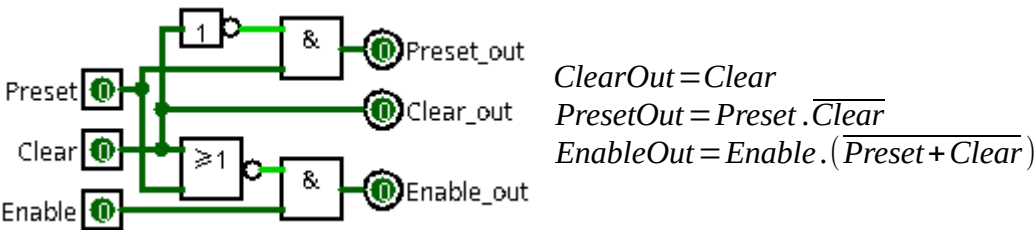
$RO=0+(RI \cdot 1)$
 $RO=0+RI$
 $RO=RI$

RI=0 : Mise à 1
RI=1 : Interdit

2-2) RSEM : (RS Enable Main)

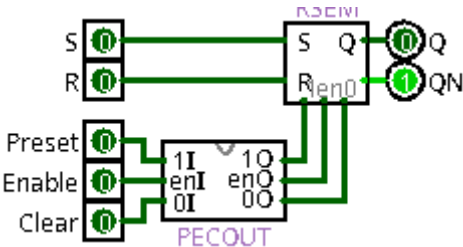


2-3) Logisim : (PECOUT)



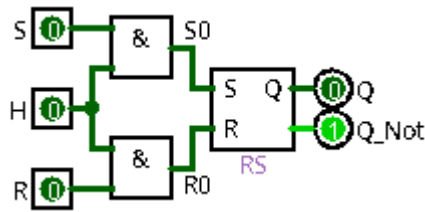
2-4) Bascule Logisim : (RSE => RS Enable)

(PECOUT / RSEM)



3) Bascule RSH :

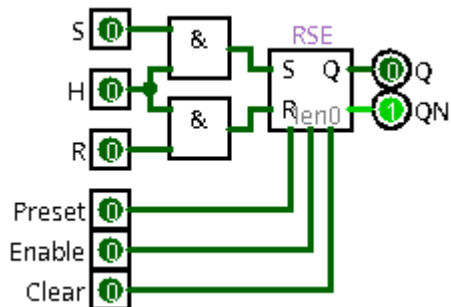
3-1) Bascule : (High Level [Niveau Haut])



Algèbre de boole

$$S0 = S \cdot H$$

$$R0 = R \cdot H$$



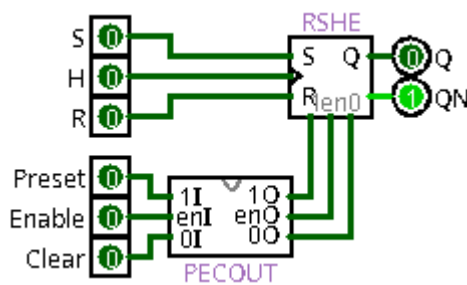
$$Q = Q_{(n-1)} \cdot (\overline{R} + \overline{H}) + S \cdot H$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{S} + \overline{H}) + R \cdot H$$

$$Q = Q_{(n-1)} \cdot \overline{R0} + S0 \quad \overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{S0} + R0$$

$$Q = Q_{(n-1)} \cdot \overline{R} \cdot \overline{H} + S \cdot H \quad \overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{S} \cdot \overline{H} + R \cdot H$$

$$Q = Q_{(n-1)} \cdot (\overline{R} + \overline{H}) + S \cdot H \quad \overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{S} + \overline{H}) + R \cdot H$$



S	R	H	Q	$\overline{Q}$	remarque
X	X	0	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	0	1	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	1	1	0	1	mise à 0
1	0	1	1	0	mise à 1
1	1	1	0	0	état interdit

H=0

$$Q = Q_{(n-1)} \cdot (\overline{R} + \overline{0}) + S \cdot 0$$

$$Q = Q_{(n-1)} \cdot (\overline{R} + 1) + S \cdot 0$$

$$Q = Q_{(n-1)} \cdot 1$$

$$Q = Q_{(n-1)}$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{S} + \overline{0}) + R \cdot 0$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{S} + 1) + R \cdot 0$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot 1$$

$$\overline{Q} = \overline{Q_{(n-1)}}$$

$$Q = Q_{(n-1)}$$

$$\overline{Q} = \overline{Q_{(n-1)}}$$

Mémoire

H=1

$$Q = Q_{(n-1)} \cdot (\overline{R} + \overline{1}) + S \cdot 1$$

$$Q = Q_{(n-1)} \cdot (\overline{R} + 0) + S \cdot 1$$

$$Q = Q_{(n-1)} \cdot \overline{R} + S$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{S} + \overline{1}) + R \cdot 1$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{S} + 0) + R \cdot 1$$

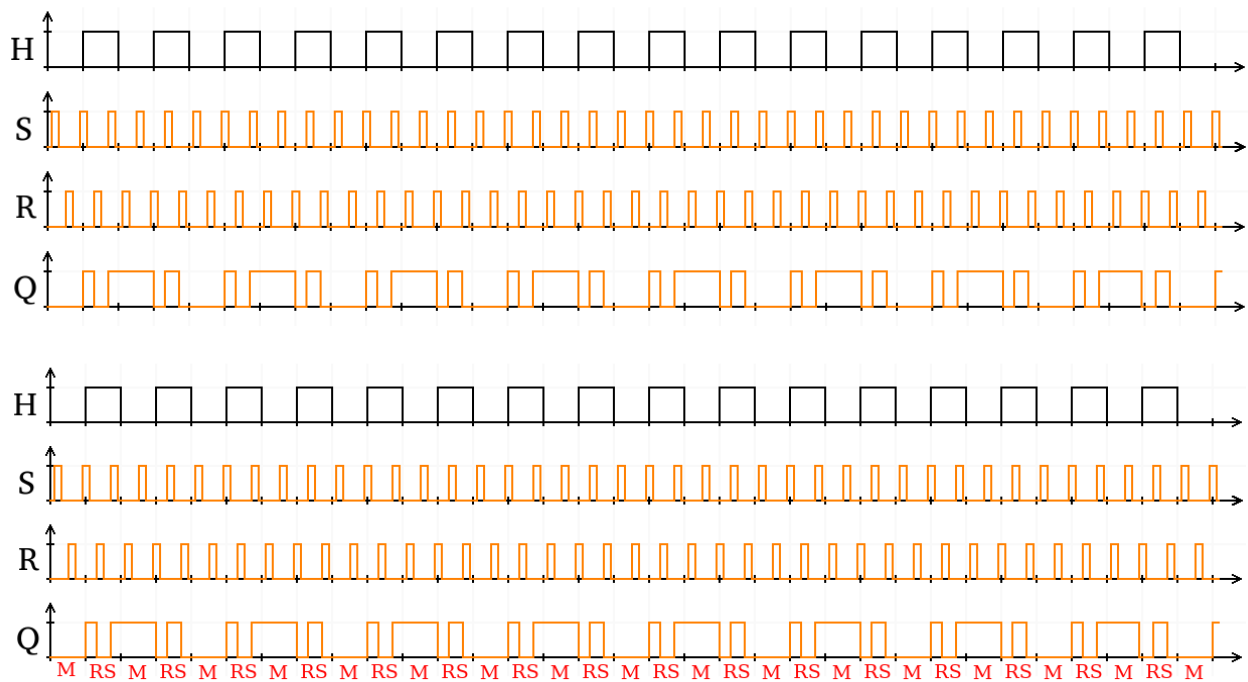
$$\overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{S} + R$$

$$Q = Q_{(n-1)} \cdot \overline{R} + S$$

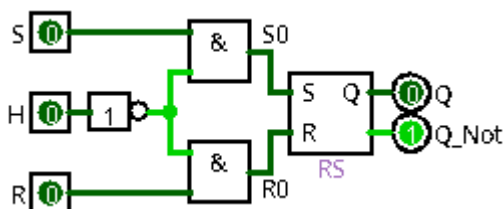
$$\overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{S} + R$$

Bascule RS

3-2) diagramme de temps :



3-3) Logisim : (RSHB => RSH Bas, Low Level [Niveau Bas])



H=1

$$Q = Q_{(n-1)}$$

$$\bar{Q} = \bar{Q}_{(n-1)}$$

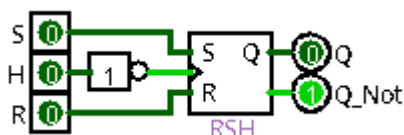
Mémoire

H=0

$$Q = Q_{(n-1)} \cdot \bar{R} + S$$

$$\bar{Q} = \bar{Q}_{(n-1)} \cdot \bar{S} + R$$

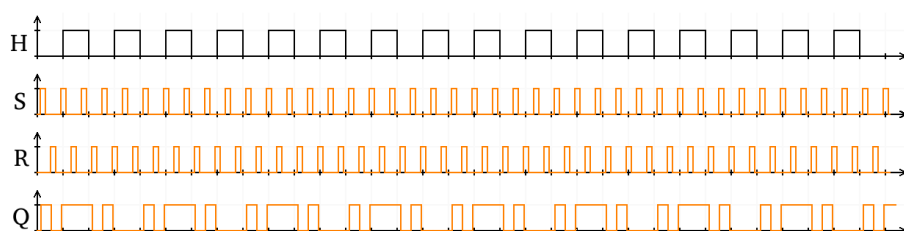
Bascule RS



$$Q = Q_{(n-1)} \cdot (\bar{R} + H) + S \cdot \bar{H}$$

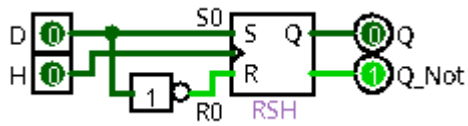
$$\bar{Q} = \bar{Q}_{(n-1)} \cdot (\bar{S} + H) + R \cdot \bar{H}$$

S	R	H	Q	$\bar{Q}$	remarque
X	X	1	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	0	0	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	1	0	0	1	mise à 0
1	0	0	1	0	mise à 1
1	1	0	0	0	état interdit

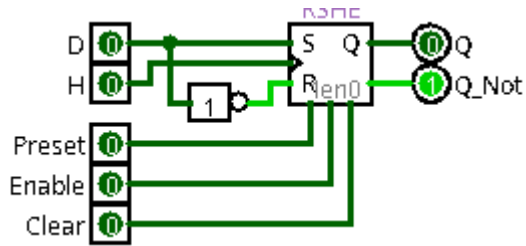


4) Bascule D : (D avec RSH)

4-1) Bascule :



(RSHE)



$$Q = Q_{(n-1)} \cdot \overline{H} + D \cdot (Q_{(n-1)} + H)$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{H} + \overline{D} \cdot (\overline{Q_{(n-1)}} + H)$$

D	H	Q	$\overline{Q}$	remarque
X	0	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	1	0	1	mise à 0
1	1	1	0	mise à 1

Algèbre de boole

$$S0 = D$$

$$R0 = \overline{D}$$

$$Q = Q_{(n-1)} \cdot (\overline{R0} + \overline{H}) + S0 \cdot H$$

$$Q = Q_{(n-1)} \cdot (\overline{D} + \overline{H}) + D \cdot H$$

$$Q = Q_{(n-1)} \cdot (D + \overline{H}) + D \cdot H$$

$$Q = Q_{(n-1)} \cdot \overline{H} + Q_{(n-1)} \cdot D + D \cdot H$$

$$Q = Q_{(n-1)} \cdot \overline{H} + D \cdot (Q_{(n-1)} + H)$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{S0} + \overline{H}) + R0 \cdot H$$

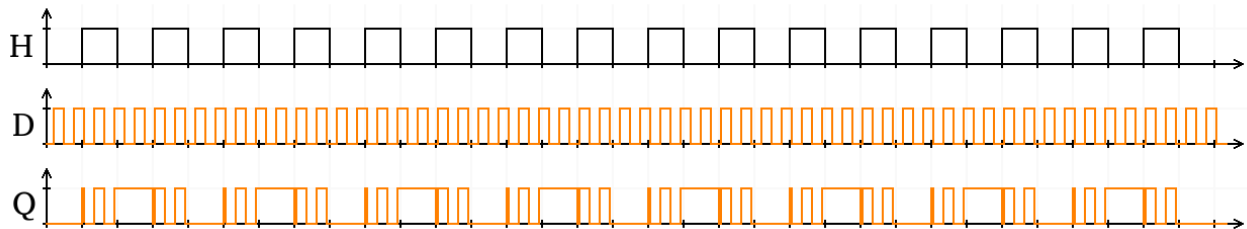
$$\overline{Q} = \overline{Q_{(n-1)}} \cdot (\overline{D} + \overline{H}) + \overline{D} \cdot H$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{D} + \overline{Q_{(n-1)}} \cdot \overline{H} + \overline{D} \cdot H$$

$$\overline{Q} = \overline{Q_{(n-1)}} \cdot \overline{H} + \overline{D} \cdot (\overline{Q_{(n-1)}} + H)$$

H=0	H=1
$Q = Q_{(n-1)} \cdot \bar{0} + D \cdot (Q_{(n-1)} + 0)$ $Q = Q_{(n-1)} \cdot 1 + D \cdot (Q_{(n-1)} + 0)$ $Q = Q_{(n-1)} + D \cdot Q_{(n-1)}$ $Q = Q_{(n-1)} \cdot (1 + D)$ $Q = Q_{(n-1)} \cdot 1$ $Q = Q_{(n-1)}$ $\bar{Q} = \overline{Q_{(n-1)}} \cdot \bar{0} + \bar{D} \cdot (\overline{Q_{(n-1)}} + 0)$ $\bar{Q} = \overline{Q_{(n-1)}} \cdot 1 + \bar{D} \cdot (\overline{Q_{(n-1)}} + 0)$ $\bar{Q} = \overline{Q_{(n-1)}} + \bar{D} \cdot \overline{Q_{(n-1)}}$ $\bar{Q} = \overline{Q_{(n-1)}} \cdot (\bar{D} + 1)$ $\bar{Q} = \overline{Q_{(n-1)}} \cdot 1$ $\bar{Q} = \overline{Q_{(n-1)}}$ $Q = Q_{(n-1)}$ $\bar{Q} = \overline{Q_{(n-1)}}$	$Q = Q_{(n-1)} \cdot \bar{1} + D \cdot (Q_{(n-1)} + 1)$ $Q = Q_{(n-1)} \cdot 0 + D \cdot (Q_{(n-1)} + 1)$ $Q = 0 + D \cdot 1$ $Q = D$ $\bar{Q} = \overline{Q_{(n-1)}} \cdot \bar{1} + \bar{D} \cdot (\overline{Q_{(n-1)}} + 1)$ $\bar{Q} = \overline{Q_{(n-1)}} \cdot 0 + \bar{D} \cdot (\overline{Q_{(n-1)}} + 1)$ $\bar{Q} = 0 + \bar{D} \cdot 1$ $\bar{Q} = \bar{D}$ $Q = D$ $\bar{Q} = \bar{D}$

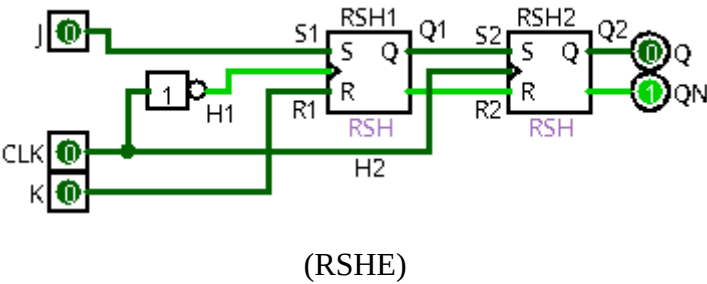
4-2) diagramme de temps :



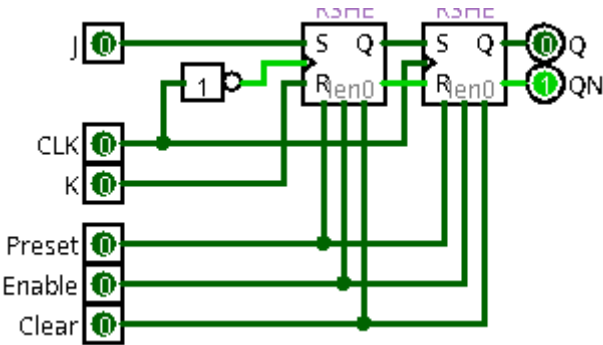
5) Bascule JK : (JKM => JK Montant)

5-1) Bascule :

Rising Edge [Front montant]



S	R	CLK	Q	Q̄	remarque
X	X	X	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	0	↑	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	1	↑	0	1	mise à 0
1	0	↑	1	0	mise à 1
1	1	↑	0	0	état interdit



5-2) RSH1 :

Algèbre de boole	CLK=0	CLK=1
$H1 = \overline{CLK}$	$Q1 = Q1_{(n-1)} \cdot (\overline{K} + 0) + J \cdot \overline{0}$	$Q1 = Q1_{(n-1)} \cdot (\overline{K} + 1) + J \cdot \overline{1}$
$Q1 = Q1_{(n-1)} \cdot (\overline{R1} + \overline{H1}) + S1 \cdot H1$	$Q1 = Q1_{(n-1)} \cdot \overline{K} + J \cdot 1$	$Q1 = Q1_{(n-1)} \cdot 1 + J \cdot 0$
$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{S1} + \overline{H1}) + R1 \cdot H1$	$Q1 = Q1_{(n-1)} \cdot \overline{K} + J$	$Q1 = Q1_{(n-1)}$
$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{CLK}) + J \cdot \overline{CLK}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + 0) + K \cdot \overline{0}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + 1) + K \cdot \overline{1}$
$Q1 = Q1_{(n-1)} \cdot (\overline{K} + CLK) + J \cdot \overline{CLK}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K \cdot 1$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot 1 + K \cdot 0$
$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{CLK}) + K \cdot \overline{CLK}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K$	$\overline{Q1} = \overline{Q1_{(n-1)}}$
$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + CLK) + K \cdot \overline{CLK}$	$Q1 = Q1_{(n-1)} \cdot \overline{K} + J$	$Q1 = Q1_{(n-1)}$
$Q1 = Q1_{(n-1)} \cdot (\overline{K} + CLK) + J \cdot \overline{CLK}$	$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K$	$\overline{Q1} = \overline{Q1_{(n-1)}}$
$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + CLK) + K \cdot \overline{CLK}$		
	Bascule RS (S=J et R=K)	Mémoire

5-3) RSH2 :

Algèbre de boole	CLK=0	CLK=1
$H1 = CLK$	$Q2 = Q2_{(n-1)} \cdot (Q1 + \overline{0}) + Q1 \cdot 0$	$Q2 = Q2_{(n-1)} \cdot (Q1 + \overline{1}) + Q1 \cdot 1$
$S2 = Q1$	$Q2 = Q2_{(n-1)} \cdot (Q1 + 1)$	$Q2 = Q2_{(n-1)} \cdot (Q1 + 0) + Q1$
$R2 = \overline{Q1}$	$Q2 = Q2_{(n-1)} \cdot 1$	$Q2 = Q2_{(n-1)} \cdot Q1 + Q1$
$Q2 = Q2_{(n-1)} \cdot (\overline{R2} + \overline{H2}) + S2 \cdot H2$	$Q2 = Q2_{(n-1)}$	$Q2 = (Q2_{(n-1)} + 1) \cdot Q1$
$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{S2} + \overline{H2}) + R2 \cdot H2$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{0}) + \overline{Q1} \cdot 0$	$Q2 = Q1$
$Q2 = Q2_{(n-1)} \cdot (\overline{Q1} + \overline{CLK}) + Q1 \cdot CLK$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 1)$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{1}) + \overline{Q1} \cdot 1$
$Q2 = Q2_{(n-1)} \cdot (Q1 + \overline{CLK}) + Q1 \cdot CLK$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot 1$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 0) + \overline{Q1}$
$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{CLK}) + \overline{Q1} \cdot CLK$	$\overline{Q2} = \overline{Q2_{(n-1)}}$	$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot Q1 + \overline{Q1}$
$Q2 = Q2_{(n-1)} \cdot (Q1 + \overline{CLK}) + Q1 \cdot CLK$	$Q2 = Q2_{(n-1)}$	$\overline{Q2} = (\overline{Q2_{(n-1)}} + 1) \cdot \overline{Q1}$
$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{CLK}) + \overline{Q1} \cdot CLK$	$\overline{Q2} = \overline{Q2_{(n-1)}}$	$\overline{Q2} = \overline{Q1}$
	Mémoire	Bascule D (D=Q1)

5-4) diagramme de temps pour RSH1 et RSH2 :**CLK=0**

$$Q1 = Q1_{(n-1)} \cdot \bar{K} + J$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \bar{J} + K$$

$$Q2 = Q2_{(n-1)}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}}$$

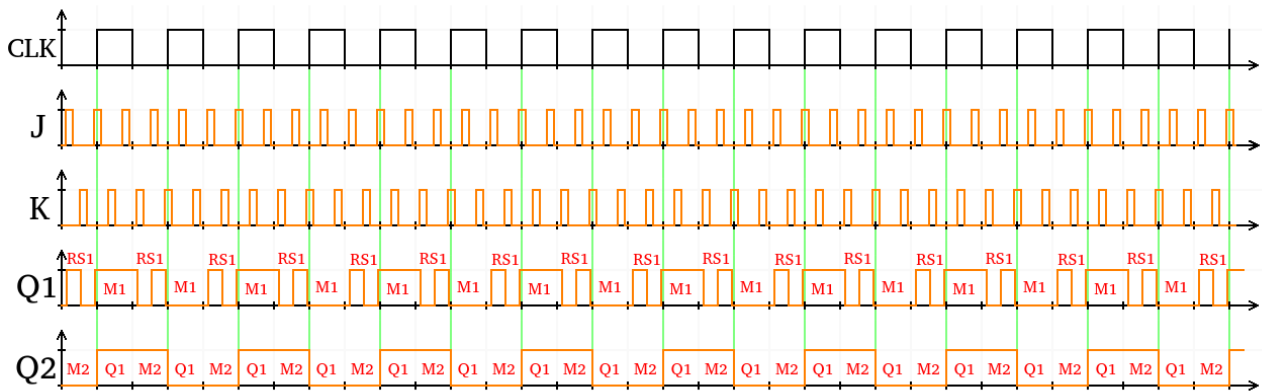
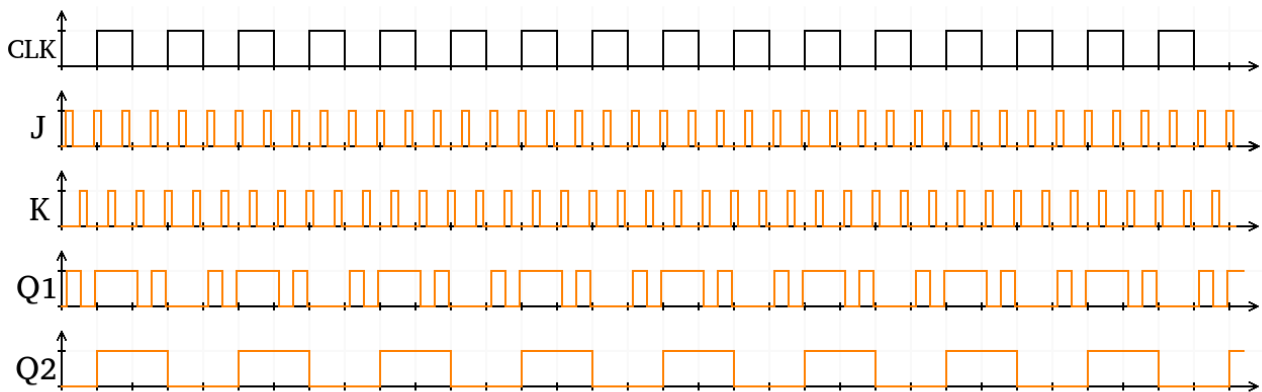
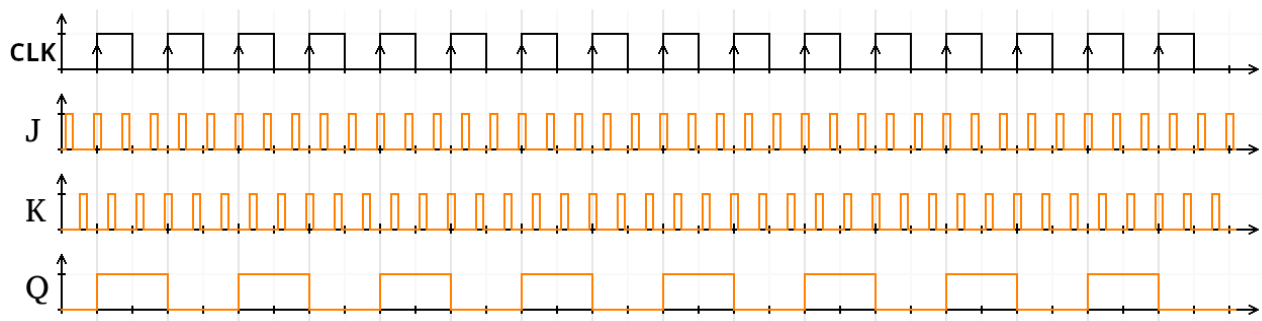
CLK=1

$$Q2 = Q1$$

$$\overline{Q2} = \overline{Q1}$$

$$Q1 = Q1_{(n-1)}$$

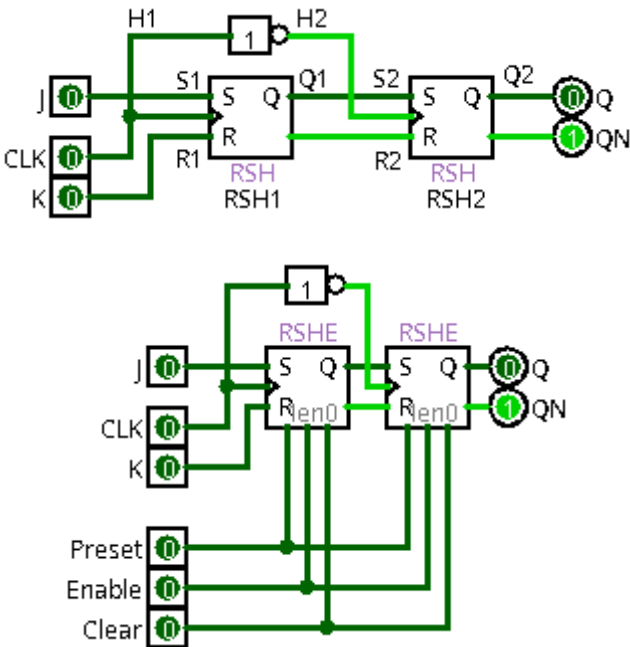
$$\overline{Q1} = \overline{Q1_{(n-1)}}$$

**5-5) diagramme de temps pour JK :**

6) Bascule JK : (JKD => JK Descendant)

6-1) Bascule :

Rising Edge [Front montant]



S	R	CLK	Q	Q̄	remarque
X	X	X	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	0	↓	$Q_{(n-1)}$	$\overline{Q_{(n-1)}}$	mémoire
0	1	↓	0	1	mise à 0
1	0	↓	1	0	mise à 1
1	1	↓	0	0	état interdit

6-2) RSH1 :**Algèbre de boole**

$$H1 = CLK$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{R1} + \overline{H1}) + S1 \cdot H1$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{S1} + \overline{H1}) + R1 \cdot H1$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{CLK}) + J \cdot CLK$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{CLK}) + K \cdot CLK$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{CLK}) + J \cdot CLK$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{CLK}) + K \cdot CLK$$

CLK=0

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{0}) + J \cdot 0$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + 1)$$

$$Q1 = Q1_{(n-1)} \cdot 1$$

$$Q1 = Q1_{(n-1)}$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{0}) + K \cdot 0$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + 1)$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot 1$$

$$\overline{Q1} = \overline{Q1_{(n-1)}}$$

$$Q1 = Q1_{(n-1)}$$

$$\overline{Q1} = \overline{Q1_{(n-1)}}$$

Mémoire

CLK=1

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + \overline{1}) + J \cdot 1$$

$$Q1 = Q1_{(n-1)} \cdot (\overline{K} + 0) + J$$

$$Q1 = Q1_{(n-1)} \cdot \overline{K} + J$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + \overline{1}) + K \cdot 1$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot (\overline{J} + 0) + K$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K$$

$$Q1 = Q1_{(n-1)} \cdot \overline{K} + J$$

$$\overline{Q1} = \overline{Q1_{(n-1)}} \cdot \overline{J} + K$$

Bascule RS (S=J et R=K)

6-3) RSH2 :

$$H1 = \overline{CLK}$$

$$S2 = Q1$$

$$R2 = \overline{Q1}$$

$$Q2 = Q2_{(n-1)} \cdot (\overline{R2} + \overline{H2}) + S2 \cdot H2$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{S2} + \overline{H2}) + R2 \cdot H2$$

$$Q2 = Q2_{(n-1)} \cdot (\overline{Q1} + \overline{CLK}) + Q1 \cdot \overline{CLK}$$

$$Q2 = Q2_{(n-1)} \cdot (Q1 + CLK) + Q1 \cdot \overline{CLK}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + \overline{CLK}) + \overline{Q1} \cdot \overline{CLK}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + CLK) + \overline{Q1} \cdot \overline{CLK}$$

$$Q2 = Q2_{(n-1)} \cdot (Q1 + CLK) + Q1 \cdot \overline{CLK}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + CLK) + \overline{Q1} \cdot \overline{CLK}$$

CLK=0

$$Q2 = Q2_{(n-1)} \cdot (Q1 + 0) + Q1 \cdot \overline{0}$$

$$Q2 = Q2_{(n-1)} \cdot Q1 + Q1 \cdot 1$$

$$Q2 = Q2_{(n-1)} \cdot Q1 + Q1$$

$$Q2 = (Q2_{(n-1)} + 1) \cdot Q1$$

$$Q2 = 1 \cdot Q1$$

$$Q2 = Q1$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 0) + \overline{Q1} \cdot \overline{0}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot \overline{Q1} + \overline{Q1} \cdot 1$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot \overline{Q1} + \overline{Q1}$$

$$\overline{Q2} = (\overline{Q2_{(n-1)}} + 1) \cdot \overline{Q1}$$

$$\overline{Q2} = 1 \cdot \overline{Q1}$$

$$\overline{Q2} = \overline{Q1}$$

$$Q2 = Q1$$

$$\overline{Q2} = \overline{Q1}$$

Bascule D (D=Q1)

CLK=1

$$Q2 = Q2_{(n-1)} \cdot (Q1 + 1) + Q1 \cdot \overline{1}$$

$$Q2 = Q2_{(n-1)} \cdot (Q1 + 1) + Q1 \cdot 0$$

$$Q2 = Q2_{(n-1)} \cdot 1 + 0$$

$$Q2 = Q2_{(n-1)}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 1) + \overline{Q1} \cdot \overline{1}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot (\overline{Q1} + 1) + \overline{Q1} \cdot 0$$

$$\overline{Q2} = \overline{Q2_{(n-1)}} \cdot 1 + 0$$

$$\overline{Q2} = \overline{Q2_{(n-1)}}$$

$$Q2 = Q2_{(n-1)}$$

$$\overline{Q2} = \overline{Q2_{(n-1)}}$$

Mémoire

6-4) diagramme de temps pour RSH1 et RSH2 :

CLK=0

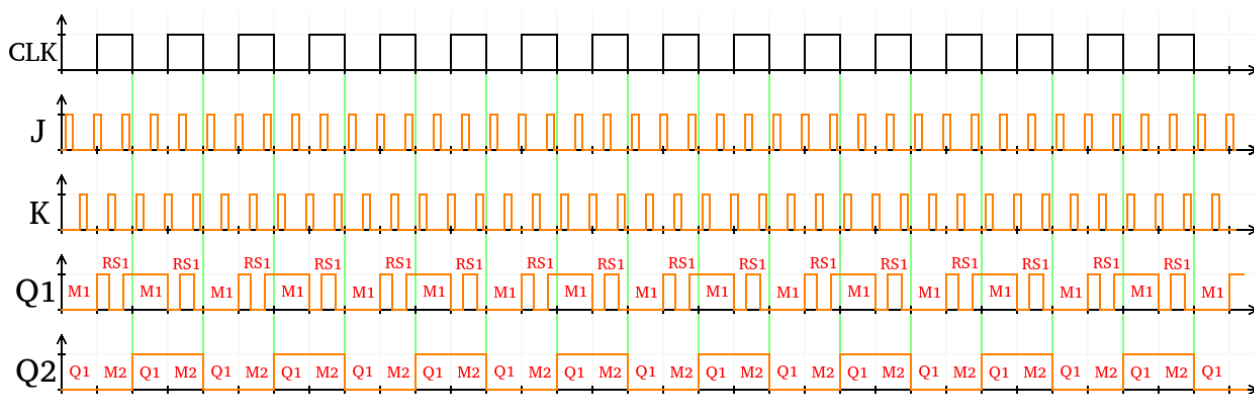
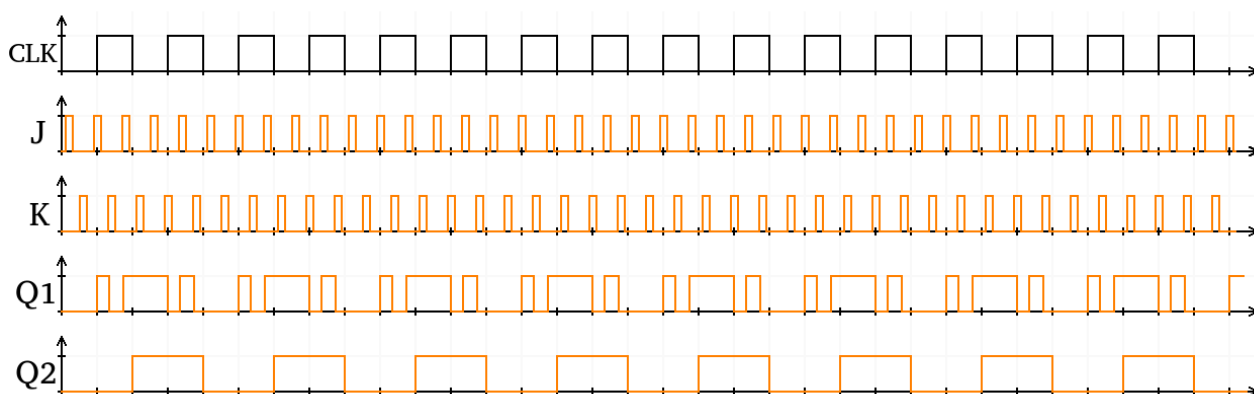
$$\frac{Q2}{Q2} = \frac{Q1}{Q1}$$

$$\frac{Q1}{Q1} = \frac{Q1_{(n-1)}}{Q1_{(n-1)}}$$

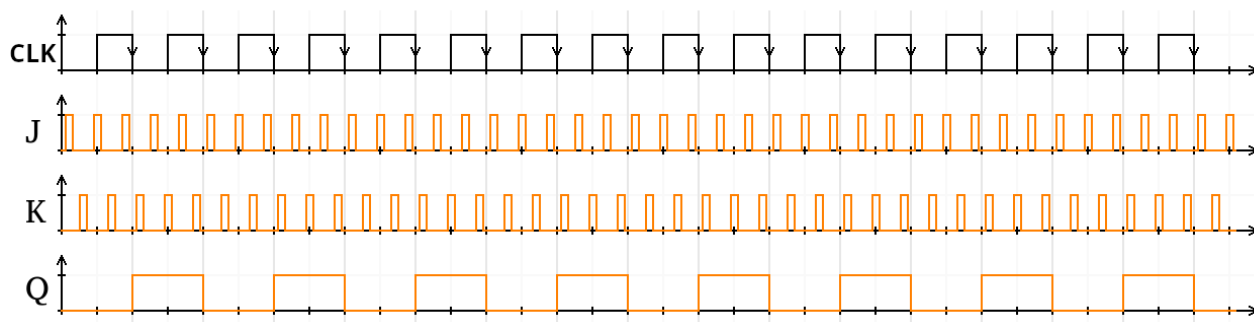
CLK=1 :

$$\frac{Q1}{Q1} = \frac{Q1_{(n-1)} \cdot \bar{K} + J}{Q1_{(n-1)} \cdot \bar{J} + K}$$

$$\frac{Q2}{Q2} = \frac{Q2_{(n-1)}}{Q2_{(n-1)}}$$

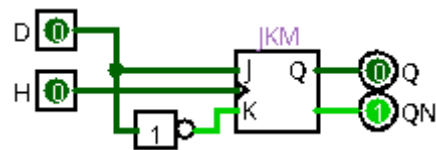
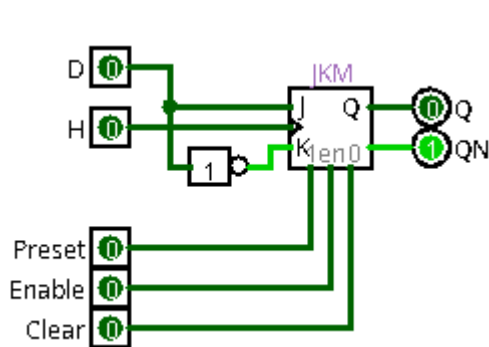


6-5) diagramme de temps pour JK :

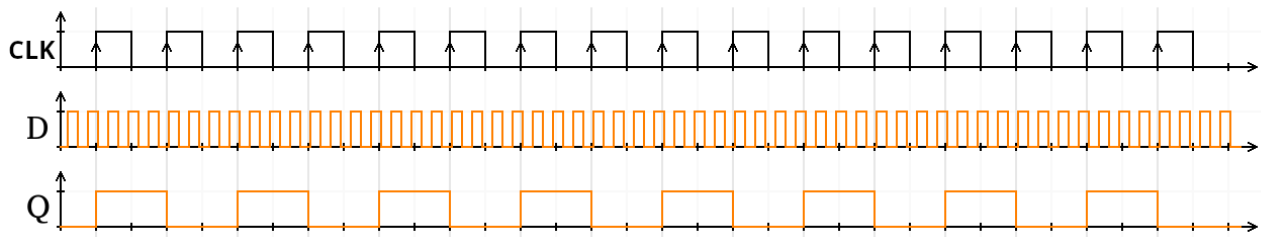


7) Bascule D : (DM => D Montant)

Rising Edge [Front montant]

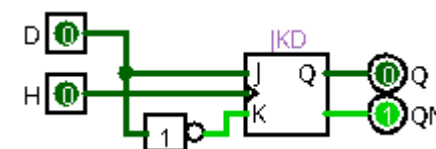
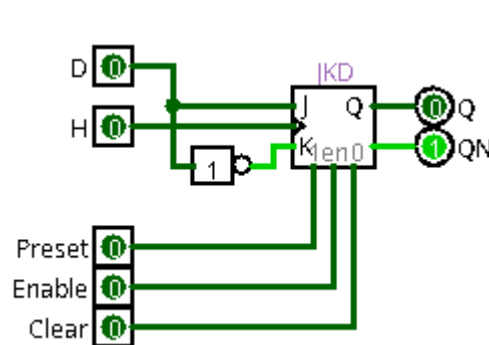


D	CLK	Q	$\bar{Q}$	remarque
X	X	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	↑	0	1	mise à 0
1	↑	1	0	mise à 1

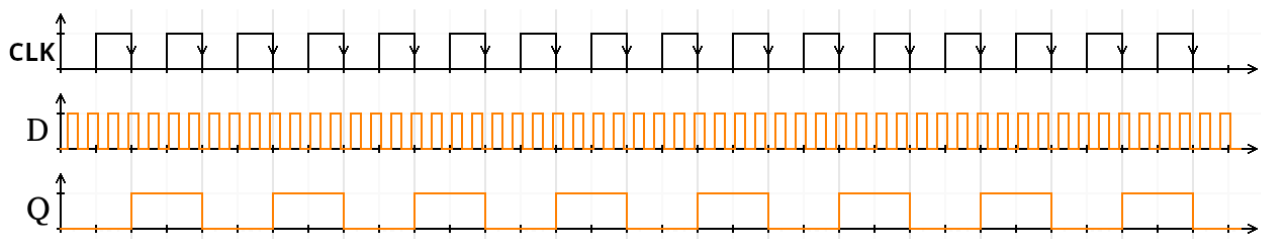


8) Bascule D : (DD => D Descendant)

Falling Edge [Front descendant]



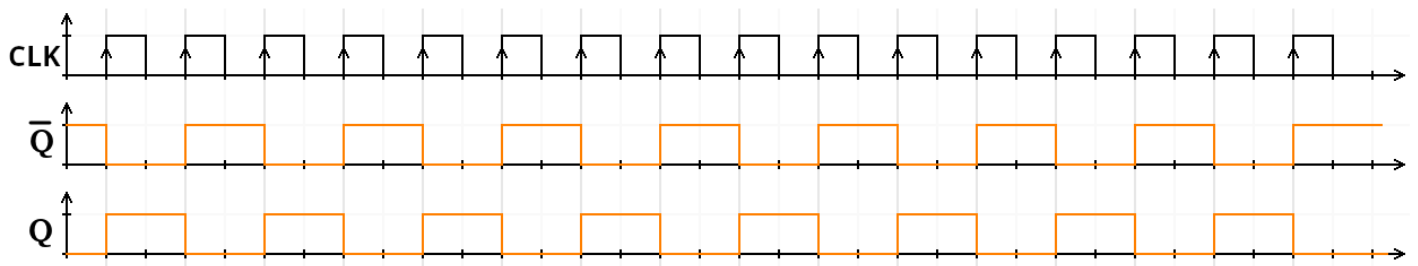
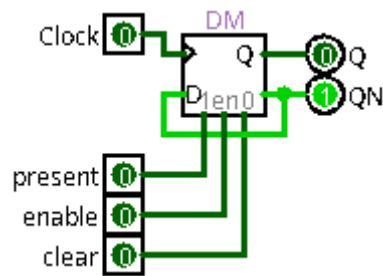
D	CLK	Q	$\bar{Q}$	remarque
X	X	$Q_{(n-1)}$	$\bar{Q}_{(n-1)}$	mémoire
0	↓	0	1	mise à 0
1	↓	1	0	mise à 1



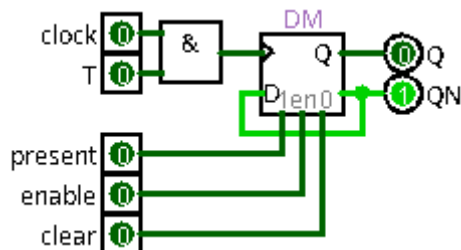
9) Bascule T Montant :

9-1) T : (TM => T Montant)

Rising Edge [Front montant]



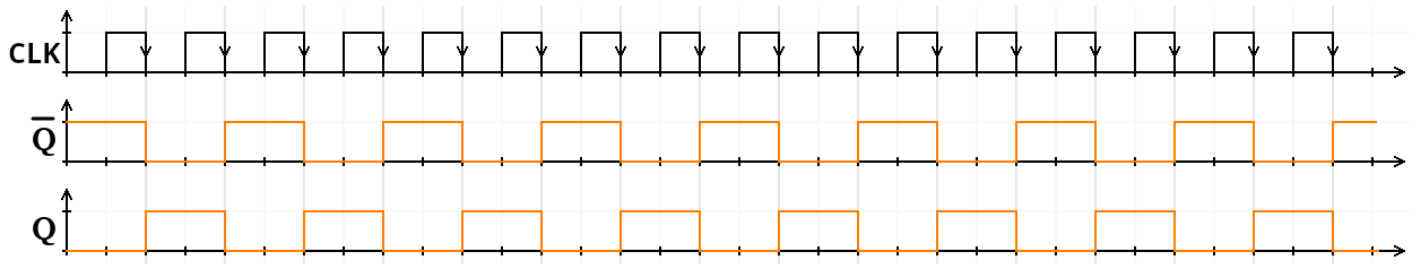
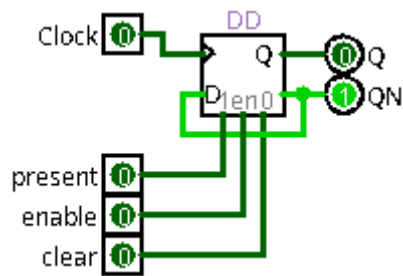
9-2) Bascule Logisim : (TLM)



10) Bascule T Descendant:

10-1) T : (TM => T Descendant)

Falling Edge [Front descendant]



10-2) Bascule Logisim : (TLD)

